ModelIt!

Teacher's Guide

Where Do Animals Live?

Animal Homes and Habitats

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: K-LS1-1

Use observations to describe patterns of what plants and animals (including humans) need to survive.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students create a model habitat that shows the key things an animal needs to survive — food, water, and shelter.
Disciplinary Core Idea	**LS1.A Structure and Function** — All organisms have external parts. Different animals use their body parts in different ways to see, hear, grasp objects, protect themselves, move from place to place, and seek, find, and take in food, water, and air.
Crosscutting Concept	**Cause and Effect** — Students observe that changing the amount of food available (cause) changes how many animals can live in a habitat (effect).

CAST Testing Alignment

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every animal on Earth lives in a habitat — a place that provides what it needs to survive. Understanding habitats helps students see how animals and their environments are connected and why protecting natural places is so important.

What Is a Habitat?

A habitat is the natural environment where an organism lives. It provides four essential things: food, water, shelter, and space. Different animals need different habitats — a fish needs water, a bird needs trees, and a polar bear needs ice and snow. When a habitat has all the resources an animal needs, the animal can survive, grow, and reproduce. When resources are scarce, animal populations shrink or animals migrate to new areas.

Food Availability and Animal Populations

The amount of food in a habitat is one of the biggest factors determining how many animals can live there. A pond full of fish can support many ducks, herons, and otters. A pond with few fish supports fewer animals. This concept is called carrying capacity. When food increases, populations can grow. When food decreases due to drought, pollution, or over-consumption, populations decline. This basic ecological principle is accessible to kindergarteners through simple scenarios about animals and their food.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Amount of Food as the external component (nature or people determine how much food is in a habitat) and Number of Animals as the internal component (it changes based on food availability).

E **Establish Relationships**
Students discover that more food leads to more animals — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Amount of Food to Number of Animals with a positive arrow.

E **Evaluate Outcomes**
Students test Little Food vs. Lots of Food scenarios and observe how the number of animals changes.

R **Revise & Extend**
Students explore what happens when shelter or water is removed from the habitat — can animals survive with food but no shelter?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Where do fish live? Where do birds live? Where do YOU live? Today we are going to learn about animal HOMES!"*

Watch For

Show pictures of animals in their habitats. Ask: Why does a fish live in water? Could a fish live in a tree? Students giggle and discuss.

Slide 2: What Will We Learn?

3 min

Say: *"Today we learn about habitats! A habitat is an animal's home — it has food, water, and a safe place to live. Every animal needs the right habitat!"*

Watch For

Introduce 'habitat' and 'shelter.' Students say habitat three times. Act it out: pretend to eat (food), drink (water), and hide under a blanket (shelter).

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: Why do different animals live in DIFFERENT places? A polar bear lives in the Arctic and a monkey lives in the jungle — why?"*

Watch For

Matching game on the board: draw a line from each animal to its habitat. Students come up one at a time to draw lines.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Amount of Food is how much food is in the habitat — nature or people control that. Number of Animals is how many animals can live there — that depends on the food!"*

Watch For

Use sorting mats. Students place Amount of Food on OUTSIDE and Number of Animals on INSIDE. Use toy animals and pretend food to demonstrate.

Slide 5: How Are They Connected?

5 min

Say: *"When there is MORE food, MORE animals can live there. When food runs out, animals have to leave. That is a POSITIVE connection!"*

Watch For

Act it out: spread toy fish around the rug (lots of food). Students pretend to be ducks and spread out. Remove most fish (little food). Now only a few ducks can stay.

Slide 6: What Did We Find Out?

5 min

Say: *"We found out that animals live in places that have what they need! More food means more animals can live there. Habitats need food, water, AND shelter."*

Watch For

Chant: 'When Food goes up, Animals go up!' Complete the sentence frame together.

Slide 7: STEM Challenge

10 min

Say: *"Now YOU are the zookeepers! Pick an animal and build it the perfect habitat. Make sure it has food, water, and shelter!"*

Watch For

Distribute shoeboxes, construction paper, craft sticks, cotton balls, plastic animals, glue. Students build habitats and present them: What animal? What food? Where is the water? Where is the shelter?

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Amount of Food
- **INTERNAL:** Number of Animals

Relationships

- **Amount of Food to Number of Animals:** POSITIVE (+) — When there is more food in a habitat, more animals can live there because every animal needs to eat to survive.

Key Discoveries

- Animals live in places that have the food, water, and shelter they need
- When there is more food in a habitat, more animals can live there
- When food is scarce, animals have to move to a new place or there are fewer of them

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals choose where to live like people choose a house"	Animals do not pick their homes the way people do. Animals live where they can find the food, water, and shelter they need to survive. Their bodies are built for their habitat — fish have gills for water, polar bears have thick fur for cold.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All animals eat the same food"	Different animals eat very different foods. Some eat plants (herbivores), some eat other animals (carnivores), and some eat both (omnivores). A rabbit eats grass and vegetables, while a hawk eats mice and rabbits.